A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 35

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids **1** Desired characteristics introduced without linked unwanted ones – faster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	2
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Bt:** reduces insecticide spraying so fewer non-target deaths, lower cost / energy **1 Roundup Ready:** weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage **1 Golden rice:** provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 4 points below. **1** Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) **1** Antibiotic-resistance marker genes could transfer to pathogenic bacteria – possible allergies – gene transfer to gut bacteria **1** Detection in the human food chain from crops approved only for animal feed **1 Economic:** seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs – herbicide-resistant crops may increase herbicide use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
01.4	1 mark for the point below. **1** Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (240 000 - 2 000) / 240 000 x 100 = 99.2% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Whales are k-selected:** slow growing, late maturing, one calf every few years, so populations recover very slowly **1** Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades **1** No regulation / open access to the ocean – whales were valuable (oil, spermaceti) and easily found **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Moratorium on commercial whaling since 1986 until populations recover **1** Total protection for certain species – whale sanctuaries **1** Limits on numbers and sizes taken – protection of suckling mothers and calves **1** Research into whale biology and threats (ship strikes, net entanglement) – exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since **1** Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses – whaling involves few nations and a single treaty body **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses **1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population **1** Deep-water aggregations easily located and caught by trawling **1** Recovery takes decades so fisheries have had to be closed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young **1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort **1** Masks the decline of the breeding population until a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Neither is a target species but both are caught in demersal trawls – skate now extinct in the Baltic and critically endangered elsewhere **1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors – many rare species with small ranges **1** High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) **1** Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches – intense competition for light drives specialisation **1** Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships – plants flower and fruit at different times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Adapted to stable conditions so they have narrow ranges of tolerance – a small change can move conditions outside the range **1** Many species are rare with small ranges / small populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) **1** Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Over half the Amazon's rainfall comes from transpiration within the forest – clearance reduces downwind rainfall **1** Rapid decay recycles nutrients so soils are shallow / poor – canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Spearman's rank correlation coefficient **1** Mann-Whitney U test **1** Chi-squared test **1** t-test **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Spearman** → a test for correlation between two sets of ranked data • **correlation** → a relationship in which the two variables change together • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chi-squared compares frequencies / counts in categories **1** Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **chi-squared** → a test comparing observed frequencies with expected frequencies • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is always some remaining uncertainty / a probability that the result was produced by chance **1** Statistics only give a level of confidence (eg 95%, 99%) that the result is significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Bars drawn above and below each mean showing plus and minus one standard deviation **1** If the bars of two means overlap a lot the difference is probably not significant – if they do not overlap it probably is **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
07.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2** Carnivores fed pellets of sandeels, herring, anchovy – overfishing sandeels reduces puffins and terns – reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) **2** High density spreads lice (which spread to wild fish) and infections – pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife – antibiotics breed resistant bacteria – reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow **2** Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors – reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) **2** Faeces and uneaten food decompose causing deoxygenation under cages – reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	8
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Predators are killed under licence to protect stock, reducing wild populations of protected / valued species **1** Better cage designs, netting, fencing, bird scarers exclude predators without killing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 5 points below. **1 Food:** cod, tuna, swordfish, sharks, turtles – extinct - dodo, great auk, passenger pigeon **1 Fashion:** leopard, snow leopard, tiger, ocelot fur – fur seals – crocodile / alligator leather – kingfisher, parrot, ostrich feathers **1 Entertainment / pets:** parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants **1 Furniture / ornaments:** mahogany, teak, ramin timber – ebony and ivory piano keys – turtle shell, shark teeth, coral and shells as souvenirs **1 Traditional medicines:** tiger (bones, claws), rhinoceros horn, seahorses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	5
09.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Threat to humans:** sharks, venomous snakes, crocodiles **1 Disease vectors:** malaria mosquitoes, tsetse flies **1 Predators of livestock:** wolves, puma, birds of prey, herons, seals **1** Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** High demand and prices make illegal trade / poaching profitable – laws are hard to enforce across borders **1** Belief in the medicine persists despite little scientific evidence – species become rarer and more valuable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **In-situ:** species remain in natural conditions with the whole community and inter-species relationships – protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • **Ex-situ:** captive breeding when in-situ is impossible or insufficient – selection criteria – difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation) – techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning) – seed banks – successes (Hawaiian goose, Iberian lynx, red kite, large blue) • **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to – both needed, in-situ fundamental **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	9
	• **plagioclimax** → a community held back from its climax by human activity **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented.	
	Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring.	
	The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is.	
	Weaker answer - Level 2 (5 marks)	
	In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together.	
	What makes the difference	
	• Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples.	
	• Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only.	
	• 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most.	
10.2	1 mark for the point below. **1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Do not write outside the box